

Racial Differences in Detection of Glaucoma Using Retinal Nerve Fiber Layer Thickness and Bruch Membrane Opening Minimum Rim Width



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- **PURPOSE:** To compare the sensitivities and specificities of the retinal nerve fiber layer thickness (RNFLT) and Bruch membrane opening minimum rim width (BMO-MRW) reference database-based criteria for detection of glaucoma in individuals of European descent (ED) and individuals of African descent (AD).
- **DESIGN:** Comparative diagnostic analysis by race
- **METHODS:** 382 eyes of 255 glaucoma patients (ED = 170, AD = 85) and 94 eyes of 50 healthy individuals (ED = 30, AD = 20) with global and sectoral RNFLT and BMO-MRW measured with Spectralis optical coherence tomography (OCT) were included. Six diagnostic criteria were evaluated: global measurement below the 5th or 1st percentile, ≥ 1 of the 6 sector measurements below the 5th or 1st percentile, and superotemporal (ST) and/or inferotemporal (IT) measurement below the 5th or 1st percentile. The sensitivities and specificities of these measurements for detection of glaucoma were compared using bootstrapping methods.
- **RESULTS:** ST and/or IT RNFLT below the 5th percentile has the best performance for detection of glaucoma among RNFLT classifications with a sensitivity (95% CI) of 89.5% (86.1, 92.5) and specificity of 87.2% (77.8, 95.1). In AD individuals, sensitivities of ST and IT RNFLT and BMO-MRW measurements below the 5th percentile criteria were lower than in ED individuals (RNFLT: 83.7% vs 92.5%, and BMO-MRW: 72.1% vs 88.5%, respectively), as well

as specificities (AD RNFLT: 73.7% and BMO-MRW: 89.5% vs ED RNFLT: 96.4% and BMO-MRW: 98.2%, respectively).

- **CONCLUSIONS:** RNFLT and BMO-MRW had consistently lower diagnostic performance in AD individuals compared with ED individuals. BMO-MRW criteria might fail to detect as many as one-third of eyes with glaucoma, specifically in AD individuals. With the current reference database, RNFLT, and especially BMO-MRW, criteria are not adequate for diagnosing glaucoma in AD individuals. (Am J Ophthalmol 2023;246: 223–235. © 2022 Elsevier Inc. All rights reserved.)

Glaucoma, a progressive optic neuropathy,¹ is the leading cause of irreversible blindness in individuals of African descent (AD).² Studies have shown that primary open angle glaucoma (POAG) disproportionately affects individuals of AD (3–4 times higher) over individuals of European descent (ED).^{3–5} Moreover, POAG begins at an earlier age^{6–11} and progresses more rapidly in patients of AD,^{11,12} resulting in greater visual impairment compared with their ED counterparts.^{11–15}

Previous clinical and histologic investigations have characterized racial differences in the ocular structures between AD and ED patients, including greater optic disc area, larger cup-to-disc ratio, deeper maximum cup depth, and thinner neuroretinal rim and retinal nerve fiber layer (RNFL) relative to the disc size in AD patients.^{16–22} These structural variations may explain at least in part the increased risk of glaucoma in the African population. For example, in the Ocular Hypertension Treatment Study, once disc size was included in the model, individuals of AD no longer had an increased risk for development of glaucoma.²³

In the current clinical practice, glaucoma is diagnosed based on several factors, including the detection of neuroretinal rim narrowing and RNFL defects, which were clinically evaluated through a dilated fundus examination, in addition to reviewing fundus photos and optical coherence tomography (OCT) scans. Although structural assessment

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of RNFL thickness (RNFLT) and Bruch membrane opening minimum rim width (BMO-MRW) using OCT have been commonly adopted in the diagnostic assessment of glaucoma,²⁴⁻²⁷ it remains unclear whether incorporating the assessment of both parameters over RNFLT alone improves our ability to detect glaucoma.

Previous reports investigating the diagnostic accuracy of RNFLT and BMO-MRW assessment in glaucoma diagnosis mainly used the area under the receiver operating characteristic (AUROC) curve analysis,²⁴⁻³⁰ with one study using both qualitative evaluation of RNFL and BMO-MRW reports and AUROC.³¹ Although the AUROC curve analysis is a useful method to compare sensitivities between RNFL and BMO-MRW, the curve distribution varies with demographics and disease severity, which may affect the clinical detection of structural abnormalities. Alternatively, RNFL and BMO-MRW abnormalities are clinically evaluated based on the OCT diagnostic classification report, which categorizes global and sectoral measurements into 3 groups: (1) within normal limits (above the 5th percentile in the reference database), (2) borderline (between the 1st and 5th percentile), and (3) outside normal limits (below the 1st percentile), which are determined by comparing the patient's values to the reference database used by a particular OCT device.^{28,32}

Although numerous studies have reported the diagnostic performance of OCT measurements of the RNFLT and BMO-MRW for glaucoma detection,^{25-27,33} few studies evaluated the differences in the diagnostic accuracy for glaucoma detection of the RNFLT and BMO-MRW based on the reference database criteria^{31,28,34} and fewer focusing on the racial differences.^{30,34} Given these differences, it is important to examine whether reference databases are accurate in classifying such diverse populations. For example, the original reference database of the Spectralis RNFL circumpapillary circle scan contains data from mainly healthy Caucasian subjects with an age range of 17-78 years. More recently, a mixed-ethnicity reference age and disc size adjusted database has been developed for optic nerve head (ONH) radial scans for evaluation of RNFL and BMO-MRW for clinical use in the United States.

The purpose of this study was to compare the glaucoma diagnostic accuracy of RNFLT and BMO-MRW based on the OCT diagnostic classification report in AD and ED individuals.

METHODS

Glaucoma patients and healthy individuals of AD and ED enrolled in the Diagnostic Innovations in Glaucoma Study (DIGS)³⁵ and the African Descent Glaucoma Evaluation Study (ADAGES) were included in this cross-sectional observational study. All research conformed to the tenets

of the Declaration of Helsinki and the Health Insurance Portability and Accountability Act and was authorized by the institutional review boards at the University of California, San Diego. Informed consent was signed by all study participants.

A total of 382 eyes of 255 glaucoma patients (ED = 170, AD = 85) and 94 eyes of 50 healthy controls (ED = 30, AD = 20) were included. All enrolled participants underwent comprehensive ophthalmic examination, including the assessment of best-corrected visual acuity, axial length (AL; IOLMaster; Carl Zeiss Meditec), slitlamp biomicroscopy, intraocular pressure (IOP; Goldmann applanation tonometry), gonioscopy, central corneal thickness (CCT; ultrasound pachymetry; DGH Technology, Inc), dilated fundus examination, and ONH stereophotography. In addition, 24-2 Swedish Interactive Thresholding Algorithm standard visual field (VF) testing (Humphrey Field Analyzer; Carl-Zeiss Meditec) and OCT (Spectralis OCT; Heidelberg Engineering) to image the neuroretinal rim and RNFL were performed on all participants. All imaging and perimetry testing were executed within a period of 6 months.

DIGS and ADAGES have overlapping protocols and inclusion and exclusion criteria, which have been previously described.³⁵ In brief, the inclusion criteria common to all participants were age ≥ 18 years, best-corrected visual acuity of 20/40 or better, and open angles evaluated using gonioscopy at the first visit. Participants were excluded if they had a history of intraocular surgery (uncomplicated cataract or glaucoma surgeries were accepted); ocular diseases including uveitis, nonglaucomatous optic neuropathy, and diabetic or hypertensive retinopathy (controlled diabetes and hypertension participants were included); trauma; any neurodegenerative diseases, including Alzheimer, Parkinson, or dementia; history of stroke; high myopia (AL > 26.0 mm); unreliable VF testing; and poor-quality Spectralis OCT scans.

Glaucoma patients were required to have glaucomatous VF damage defined as having at least 2 consecutive and reliable (fixation losses, false positives [FPs], and false negatives [FNs] $\leq 33\%$) abnormal Humphrey VF tests with either a pattern standard deviation (PSD) outside the 95% normal limits or a Glaucoma Hemifield Test (GHT) result outside the 99% normal limit.

Healthy participants were defined as having normal-appearing optic discs, neuroretinal rims, IOP ≤ 21 mm Hg without any history of elevated IOP, and at least 2 reliable normal 24-2 VF results (PSD within 95% confidence limits and GHT results within normal limits in either or both eyes).

• **OCT IMAGING:** The optic disc and circumpapillary RNFL were imaged using optic nerve head-radial and circle (ONH-RC) scan pattern of Spectralis OCT Glaucoma Module Premium Edition (GMPE) software (Spectralis [Heidelberg Engineering Inc] software version used in

the analysis). ONH-RC scans, each composed of 24 equally spaced radial B-scans centered over the ONH, with 768 A-scans per B-scan were acquired for the measurements of BMO-MRW. RNFL was computed from the 12-degree 3.5-mm-diameter circle scan of the ONH-RC. The Imaging Data Evaluation and Analysis (IDEA) Reading Center at the University of California, San Diego, completed the quality review of all OCT scans. Only good-quality images with proper centering, focus, segmentation, and quality score were included in the analysis. The BMO, internal limiting membrane, and RNFL segmentation on the circle scan were reviewed and manually corrected, when needed, by the IDEA Reading Center.³⁶

• **RNFLT AND BMO-MRW DIAGNOSTIC CLASSIFICATION FOR GLAUCOMA DETECTION:** The most recent reference database in the US version of the Spectralis GMPE software to measure RNFLT and BMO-MRW includes 368 normal participants (20-90 years old) of various racial and ethnic descents with adjustments for age and optic disc size. Of those, 246 participants are Caucasians (66.8%) and 47 participants (12.8%) are from AD. Global and sectoral (superotemporal [ST], superonasal [SN], nasal [N], inferonasal [IN], inferotemporal [IT], and temporal [T]) RNFLT and BMO-MRW measurements provided by the Spectralis OCT were included in the analysis. These measurements were compared to the OCT reference database and classified into normal range (within normal limits), between the 1st and 5th percentile (borderline) or below the 1st percentile of the normal reference values (outside normal limits). The following 6 diagnostic criteria were evaluated to compare the sensitivities and specificities of RNFLT and BMO-MRW for the detection of glaucoma in all participants of AD and ED:

1. Global measurement below the 5th percentile,
2. Global measurement below the 1st percentile,
3. One or more sectoral measures below the 5th percentile
4. One or more sectoral measures below the 1st percentile
5. Superotemporal and/or inferotemporal measurement below the 5th percentile
6. Superotemporal and/or inferotemporal measurement below the 1st percentile

Sensitivity was defined as the proportion of glaucomatous eyes having the correct diagnostic criteria of glaucoma and specificity as the proportion of healthy eyes not meeting the diagnostic criteria of glaucoma. In addition to calculating the sensitivities and specificities of the RNFLT or BMO-MRW alone, the sensitivities and specificities were also estimated for the combination between the parameters.

• **STATISTICAL ANALYSIS:** Demographic and clinical characteristic data were presented as mean (95% CI) for continuous variables and count (%) for categorical variables. Categorical variables were compared using the Fisher exact test. Eye characteristics were compared using linear mixed

effects models to account for within-participant variability. The sensitivities and specificities of RNFLT and BMO-MRW for the detection of glaucoma in ED and AD participants were compared using a bootstrapping resampling method. In addition, sensitivity and specificity were evaluated separately for the group of patients with early glaucoma (VF MD > -6 dB).³⁷ *P* values <.05 were considered statistically significant. All statistical analyses were performed using the R statistical software (version 3.5.2).

RESULTS

A total of 476 eyes of 305 participants consisting of 382 eyes of 255 glaucoma patients and 94 eyes of 50 healthy controls were included. The participants were stratified by race into 85 glaucoma patients of AD, 170 glaucoma patients of ED, 20 healthy participants of AD, and 30 healthy subjects of ED. The demographic and clinical characteristics by diagnosis (stratified by race) are presented in Table 1. Glaucoma patients of ED were significantly older (74.5 [95% CI 73.0, 75.9] years) compared with AD glaucoma patients (69.6 [95% CI 67.2, 72.0] years) (*P* < .001), and had significantly less hypertension and diabetes.

Glaucoma patients of AD had a larger BMO area (2.32 [95% CI 2.22, 2.42] mm²) and higher IOP (14.8 [95% CI 14.0, 15.6] mm Hg) compared with ED glaucoma patients (1.95 [95% CI 1.88, 2.02] mm² and 13.1 [95% CI 12.4, 13.7] mm Hg, respectively) (*P* < .001 for all). VF MD, global BMO-MRW, RNFLT, AL, and CCT were all similar among glaucoma patients of both descents (*P* ≥ .05). In healthy eyes, mean Global BMO-MRW was larger in ED compared with AD individuals (345.5 [95% CI 322.7, 368.3] μm and 310.8 [95% CI 282.9, 338.8] μm, respectively), but the difference did not reach statistical significance (*P* = .066).

With the exception of more hypertension in healthy AD compared with ED individuals, none of the other clinical parameters were significantly different between the healthy participants (*P* > .1). Glaucoma patients of both descents had lower VF MD, smaller global BMO-MRW, and thinner RNFL than healthy controls (*P* < .001 for all).

• **SENSITIVITIES AND SPECIFICITIES OF RNFLT AND BMO-MRW FOR DETECTION OF GLAUCOMA IN ALL PARTICIPANTS:** Sensitivities of RNFLT were higher than BMO-MRW for all 6 diagnostic criteria examined (*P* < .05 for all). Conversely, specificities of BMO-MRW tended to be higher than specificities of RNFLT (Table 2). Among all 6 diagnostic criteria, ST and/or IT RNFLT measurement below the 5th percentile yielded the best performance to detect all severities of glaucoma with a sensitivity of 89.5% (95% CI 86.1, 92.5), which was significantly higher than that of ST and/or IT BMO-MRW measurement below the 5th percentile (83.0% [95% CI 78.2, 87.3], *P* = .004). However, ST and/or IT RNFLT measurement below the 5th per-

TABLE 1. Patient and Clinical Characteristics by Diagnosis (Stratified by Race)

	Glaucoma			Healthy		
	African (n = 85, Eyes = 129)	European (n = 170, Eyes = 253)	P Value	African (n = 20, Eyes = 38)	European (n = 30, Eyes = 56)	P Value
Age	69.6 (67.2, 72.0)	74.5 (73.0, 75.9)	<.001	63.0 (55.5, 70.6)	58.0 (50.9, 65.2)	.325
Gender						
Female	44 (51.8%)	91 (53.5%)	.792	15 (75.0%)	23 (76.7%)	>.999
Male	41 (48.2%)	79 (46.5%)		5 (25.0%)	7 (23.3%)	
Self-reported hypertension						
No	20 (23.5%)	80 (47.1%)	<.001	7 (35.0%)	22 (73.3%)	.010
Yes	65 (76.5%)	90 (52.9%)		13 (65.0%)	8 (26.7%)	
Self-reported diabetes						
No	48 (56.5%)	156 (91.8%)	<.001	15 (75.0%)	28 (93.3%)	.100
Yes	37 (43.5%)	14 (8.2%)		5 (25.0%)	2 (6.7%)	
VF MD (dB)	−8.79 (−10.17, −7.40)	−7.35 (−8.34, −6.37)	.099	−0.38 (−0.88, 0.11)	−0.40 (−0.81, 0.01)	.949
Global BMO-MRW (μm)	184.4 (173.3, 195.6)	181.9 (174.0, 189.8)	.717	310.8 (282.9, 338.8)	345.5 (322.7, 368.3)	.066
RNFLT (μm)	68.0 (65.0, 71.1)	67.9 (65.8, 70.1)	.957	99.6 (95.1, 104.1)	99.8 (96.1, 103.5)	.964
BMO area (mm ²)	2.32 (2.22, 2.42)	1.95 (1.88, 2.02)	<.001	2.10 (1.90, 2.30)	1.91 (1.75, 2.07)	.156
AL (mm)	24.0 (23.8, 24.3)	24.0 (23.9, 24.2)	.971	23.8 (23.3, 24.2)	23.9 (23.5, 24.2)	.742
IOP (mm Hg)	14.8 (14.0, 15.6)	13.1 (12.4, 13.7)	.001	13.8 (12.8, 14.8)	13.8 (13.0, 14.7)	.958
CCT (μm)	532.6 (523.4, 541.9)	534.9 (528.3, 541.9)	.696	538.5 (519.8, 557.2)	541.5 (526.0, 557.1)	.809

AL= axial length, BMO = Bruch membrane opening, CCT = central corneal thickness, IOP = intraocular pressure, MD = mean deviation, MRW = minimum rim width, RNFLT = retinal nerve fiber layer thickness, VF = visual field.

Statistical significance: $P < .05$.

TABLE 2. Sensitivities and Specificities of RNFLT and BMO-MRW for Detection of Glaucoma in All Participants Irrespective of Race

	RNFLT	BMO-MRW	Combined RNFLT or BMO-MRW	Difference (RNFLT-BMO MRW)	P Value*
1. Global measurement below the 5th percentile					
Sensitivity	77.7 (72.8, 82.2)	67.0 (61.4, 72.1)	83.0 (78.1, 86.9)	10.7 (6.0, 16.2)	<.001
Specificity	95.7 (90.8, 99.0)	98.9 (95.8, 100.0)	95.7 (90.8, 99.0)	−3.2 (−7.7, 0.0)	.080
2. Global measurement below the 1st percentile					
Sensitivity	64.9 (59.7, 70.5)	43.5 (37.9, 49.2)	68.3 (63.0, 73.7)	21.5 (16.3, 27.2)	<.001
Specificity	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—
3. One or more sectoral measures below the 5th percentile					
Sensitivity	91.6 (88.7, 94.5)	86.6 (82.4, 90.5)	95.5 (93.4, 97.5)	5.0 (0.8, 9.2)	.023
Specificity	77.7 (66.7, 88.2)	91.5 (83.9, 97.3)	74.5 (63.0, 85.5)	−13.8 (−23.9, −4.3)	.007
4. One or more sectoral measures below the 1st percentile					
Sensitivity	83.2 (79.0, 86.6)	74.6 (69.4, 80.2)	89.3 (85.6, 92.5)	8.6 (3.6, 13.8)	<.001
Specificity	92.6 (84.6, 98.3)	98.9 (95.8, 100.0)	92.6 (84.6, 98.3)	−6.4 (−13.9, −2.0)	.029
5. Superotemporal and/or inferotemporal measurement below the 5th percentile					
Sensitivity	89.5 (86.1, 92.5)	83.0 (78.2, 87.3)	92.7 (89.4, 95.1)	6.5 (2.3, 10.9)	.004
Specificity	87.2 (77.8, 95.1)	94.7 (87.0, 100.0)	85.1 (75.3, 94.2)	−7.4 (−16.3, 0.0)	.079
6. Superotemporal and/or inferotemporal measurement below the 1st percentile					
Sensitivity	82.2 (78.0, 85.9)	67.3 (62.0, 72.6)	85.9 (82.1, 89.6)	14.9 (10.4, 20.4)	<.001
Specificity	96.8 (90.6, 100.0)	100.0 (100.0, 100.0)	96.8 (90.6, 100.0)	−3.2 (−9.4, 0.0)	.170

AD = African descent, BMO-MRW = Bruch membrane opening minimum rim width, ED = European Descent, RNFLT = retinal nerve fiber layer thickness.

Estimates of sensitivity and specificity (%) of RNFLT and BMO-MRW (95% CI) for all glaucoma patients vs healthy controls for each diagnostic criteria including all participants.

*P value for difference between RNFLT and BMO-MRW sensitivity and specificity. Statistical significance: $P < .05$.

TABLE 3. Sensitivities and Specificities of RNFLT and BMO-MRW for Detection of Glaucoma in AD and ED Participants

	Race	RNFLT	BMO-MRW	Combined RNFLT or BMO-MRW	Difference (RNFLT-BMO MRW)	P Value*
1. Global measurement below the 5th percentile						
Sensitivity	AD	77.5 (69.7, 86.3)	62.0 (50.6, 71.5)	82.2 (74.2, 90.0)	15.5 (6.0, 26.2)	.003
	ED	77.9 (71.6, 82.7)	69.6 (62.2, 75.1)	83.4 (77.7, 87.9)	8.3 (3.4, 14.8)	.005
Specificity	AD	94.7 (87.2, 100.0)	97.4 (90.2, 100.0)	94.7 (87.2, 100.0)	-2.6 (-11.1, 0.0)	.326
	ED	96.4 (90.4, 100.0)	100.0 (100.0, 100.0)	96.4 (90.4, 100.0)	-3.6 (-9.6, 0.0)	.155
2. Global measurement below the 1st percentile						
Sensitivity	AD	67.4 (57.7, 77.7)	36.4 (26.9, 46.5)	69.0 (59.5, 79.0)	31.0 (20.8, 42.7)	<.001
	ED	63.6 (57.0, 69.8)	47.0 (40.2, 53.6)	68.0 (61.6, 74.4)	16.6 (10.8, 23.6)	<.001
Specificity	AD	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—
	ED	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—
3. One or more sectoral measures below the 5th percentile						
Sensitivity	AD	87.6 (81.3, 93.8)	76.7 (67.2, 84.6)	92.2 (87.0, 96.8)	10.9 (1.8, 20.1)	.020
	ED	93.7 (90.3, 96.3)	91.7 (87.4, 95.0)	97.2 (95.0, 99.1)	2.0 (-2.5, 6.0)	.358
Specificity	AD	65.8 (45.9, 85.0)	86.8 (71.8, 100.0)	63.2 (42.9, 80.6)	-21.1 (-40.0, -3.2)	.029
	ED	85.7 (73.2, 94.1)	94.6 (87.7, 100.0)	82.1 (69.3, 92.4)	-8.9 (-22.0, 0.00)	.097
4. One or more sectoral measures below the 1st percentile						
Sensitivity	AD	80.6 (73.0, 88.0)	62.8 (51.8, 73.0)	85.3 (77.8, 92.3)	17.8 (8.3, 29.6)	<.001
	ED	84.6 (79.6, 88.7)	80.6 (74.9, 86.3)	91.3 (86.9, 94.9)	4.0 (-1.2, 9.4)	.143
Specificity	AD	84.2 (66.7, 96.9)	97.4 (90.2, 100.0)	84.2 (66.7, 96.9)	-13.2 (-28.2, -2.7)	.038
	ED	98.2 (92.4, 100.0)	100.0 (100.0, 100.0)	98.2 (92.4, 100.0)	-1.8 (-7.6, 0.0)	.350
5. Superotemporal and/or inferotemporal measurement below the 5th percentile						
Sensitivity	AD	83.7 (76.0, 90.5)	72.1 (61.5, 80.8)	87.6 (80.7, 93.6)	11.6 (2.4, 21.1)	.017
	ED	92.5 (89.1, 95.8)	88.5 (83.6, 92.7)	95.3 (92.1, 97.6)	4.0 (-0.3, 8.2)	.068
Specificity	AD	73.7 (53.4, 91.4)	89.5 (74.4, 100.0)	71.1 (50.0, 90.0)	-15.8 (-38.0, 0.0)	.094
	ED	96.4 (90.0, 100.0)	98.2 (93.2, 100.0)	94.6 (87.6, 100.0)	-1.8 (-8.5, 4.2)	.574
6. Superotemporal and/or inferotemporal measurement below the 1st percentile						
Sensitivity	AD	79.8 (72.3, 87.8)	55.8 (45.8, 66.8)	81.4 (73.8, 88.7)	24.0 (14.9, 35.6)	<.001
	ED	83.4 (78.4, 87.7)	73.1 (67.1, 79.4)	88.1 (83.8, 92.2)	10.3 (4.9, 15.6)	<.001
Specificity	AD	92.1 (76.0, 100.0)	100 (100.0, 100.0)	92.1 (76.0, 100.0)	-7.9 (-24.0, -0.0)	.173
	ED	100.0 (100.0, 100.0)	100 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—

AD = African descent, BMO-MRW = Bruch membrane opening minimum rim width, ED = European descent, RNFLT = retinal nerve fiber layer thickness.

Estimates of sensitivity and specificity (%) of RNFLT and BMO-MRW (95% CI) for all glaucoma patients vs healthy controls for each diagnostic criteria including AD and ED participants.

*P value for difference between RNFLT and BMO-MRW sensitivity and specificity. Statistical significance: $P < .05$.

centile had a trend of lower specificity compared to the same criteria of BMO-MRW (87.2% [95% CI 77.8, 95.1] and 94.7% [95% CI 87.0, 100], $P = .079$), respectively (Table 2). For the same diagnostic criterion, the FP rate was higher for RNFLT (12.8%) than that for BMO-MRW (5.3%).

• **SENSITIVITIES AND SPECIFICITIES OF RNFLT AND BMO-MRW FOR DETECTION OF GLAUCOMA IN AD AND ED PARTICIPANTS:** Table 3 and Figure 1 compare the sensitivities and specificities of RNFLT and BMO-MRW in partic-

ipants of AD and ED. Regardless of race, sensitivities of the RNFLT were higher and specificities were lower than those of BMO-MRW in all 6 diagnostic criteria. ST and/or IT RNFLT below the 5th percentile maintained the best performance for detection of glaucoma in both groups. In participants of AD, sensitivities of ST and/or IT RNFLT and BMO-MRW measurements below the 5th percentile criteria were significantly lower than in ED participants (RNFLT: 83.7% [95% CI 76.0, 90.5] vs 92.5% [95% CI 89.1, 95.8] and BMO-MRW: 72.1% [95% CI 61.5, 80.8] vs 88.5% [95% CI 83.6, 92.7], respectively, $P = .035$ and $.002$), with

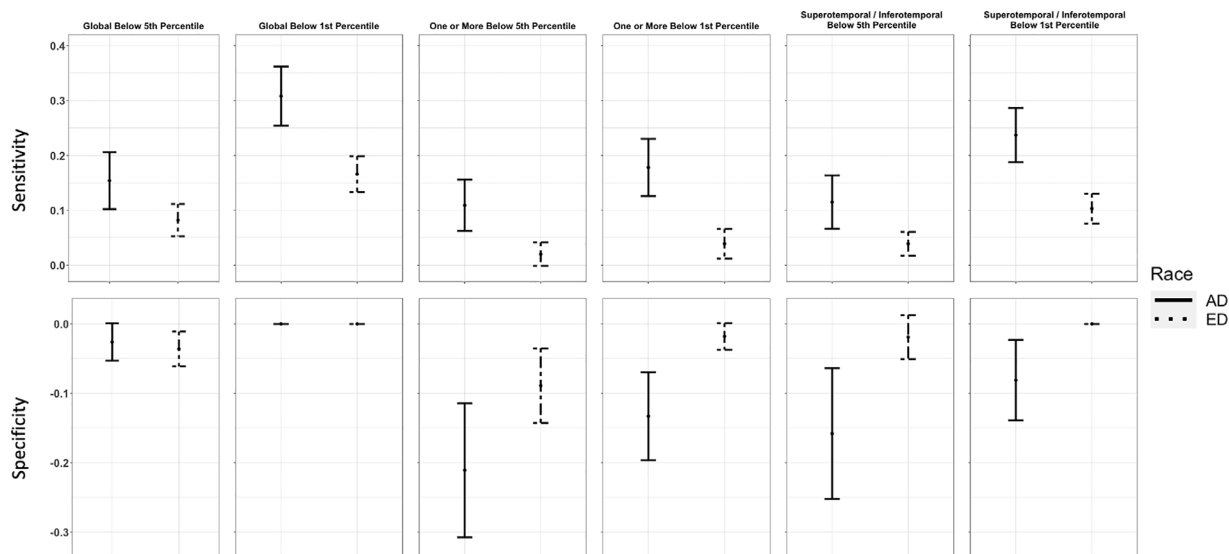


FIGURE 1. Sensitivities and specificities of RNFLT and BMO-MRW for detection of glaucoma in AD and ED participants. Error bars represent the 95% CI. AD = African descent, BMO-MRW = Bruch membrane opening minimum rim width, ED = European descent, RNFLT = retinal nerve fiber layer thickness.

lower specificities in AD participants (AD [RNFLT and BMO-MRW]: 73.7% [95% CI 53.4, 91.4] and 89.5% [74.4, 100.0] vs ED [RNFLT and BMO-MRW]: 96.4% [95% CI 90.0, 100.0] and 98.2% [95% CI 93.2, 100.0], respectively, $P = .020$ and $.235$).

The rate of FPs was consistently higher for RNFL and BMO-MRW measurements in AD (RNFL: 26.3%, BMO-MRW: 10.5%) compared to ED (RNFL: 3.6%, BMO-MRW: 1.8%). In AD glaucomatous patients, 23.4% of eyes had completely normal BMO-MRW based on the diagnostic classification report compared to 12.4% for an RNFLT diagnostic report. [Figure 2](#) represents a glaucomatous eye of an AD patient with abnormal RNFLT and glaucomatous visual field defect, but within normal limit BMO-MRW printout in all sectors. [Figure 3](#) represents a healthy eye of an AD patient with normal visual field results, but below the 5th percentile global and ST sector BMO-MRW and RNFLT (yellow) and below 1st percentile BMO-MRW and RNFLT in the T sector (red).

• **SENSITIVITIES AND SPECIFICITIES OF RNFLT AND BMO-MRW FOR DETECTION OF EARLY GLAUCOMA IN AD AND ED PARTICIPANTS:** The analysis was repeated to include 255 eyes with early glaucoma (mean VF MD: -2.04 ± 1.95 dB) using the same diagnostic criteria in AD and ED participants ([Table 4](#) and [Figure 4](#)). Similar to the results of all glaucomatous eyes, the ST and/or IT RNFLT below the 5th percentile showed the highest sensitivity and lowest specificity for detection of early glaucoma in both descents. In AD participants, sensitivities of ST and IT RNFLT and BMO-MRW measurements below the 5th percentile criteria to detect mild glaucoma were significantly lower than in ED participants (RNFL: 71.4% [95% CI 59.1, 82.8] vs

87.3% [95% CI 81.5, 92.9] and BMO-MRW: 57.1% [95% CI 43.1, 68.9] vs 82.1% [95% CI 74.6, 88.8]), respectively, with lower specificities in AD participants (AD [RNFLT and BMO-MRW]: 73.7% [95% CI 53.4, 91.4] and 89.5% [95% CI 74.4, 100.0] vs ED [RNFLT and BMO-MRW]: 96.4% [95% CI 90.0, 100.0] and 98.2% [95% CI 93.2, 100.0]), respectively.

• **COMBINING RNFLT AND BMO-MRW ASSESSMENT FOR THE DETECTION OF GLAUCOMA:** Integrating the combination of RNFL or BMO-MRW measurements with the reference to the diagnostic criterion ST and/or IT RNFLT below the 5th percentile in all participants significantly increased the sensitivities for RNFLT (all: 89.5% to 92.7%, $P < .001$; early: 81.9% to 87.7%, $P < .001$) and BMO-MRW assessment (all: 83.0% to 92.7%, $P < .001$; early: 73.5% to 87.7%, $P < .001$) without compromising their specificities for the detection of early glaucoma (RNFL: $P = .258$, and BMO-MRW: $P = .063$).

For AD participants, integrating the combination of RNFL or BMO-MRW measurements with the reference to the diagnostic criterion ST and/or IT RNFLT below the 5th percentile increased the sensitivity of RNFLT assessment without compromising its specificity for any glaucoma (sensitivity: 83.7% to 87.6%, $P = .004$; specificity: 73.7% to 71.1%, $P = .248$) and early glaucoma (sensitivity: 71.4% to 78.6%, $P < .001$; specificity: 73.7% to 71.1%, $P = .387$). Although the sensitivity of BMO-MRW assessment increased (all: 72.1% to 87.6%, $P < .001$; early: 57.1% to 78.6%, $P < .001$), its specificity decreased for any glaucoma (89.5% to 71.1%, $P = .001$) and early glaucoma (89.5% to 71.1%, $P = .013$) in AD participants.

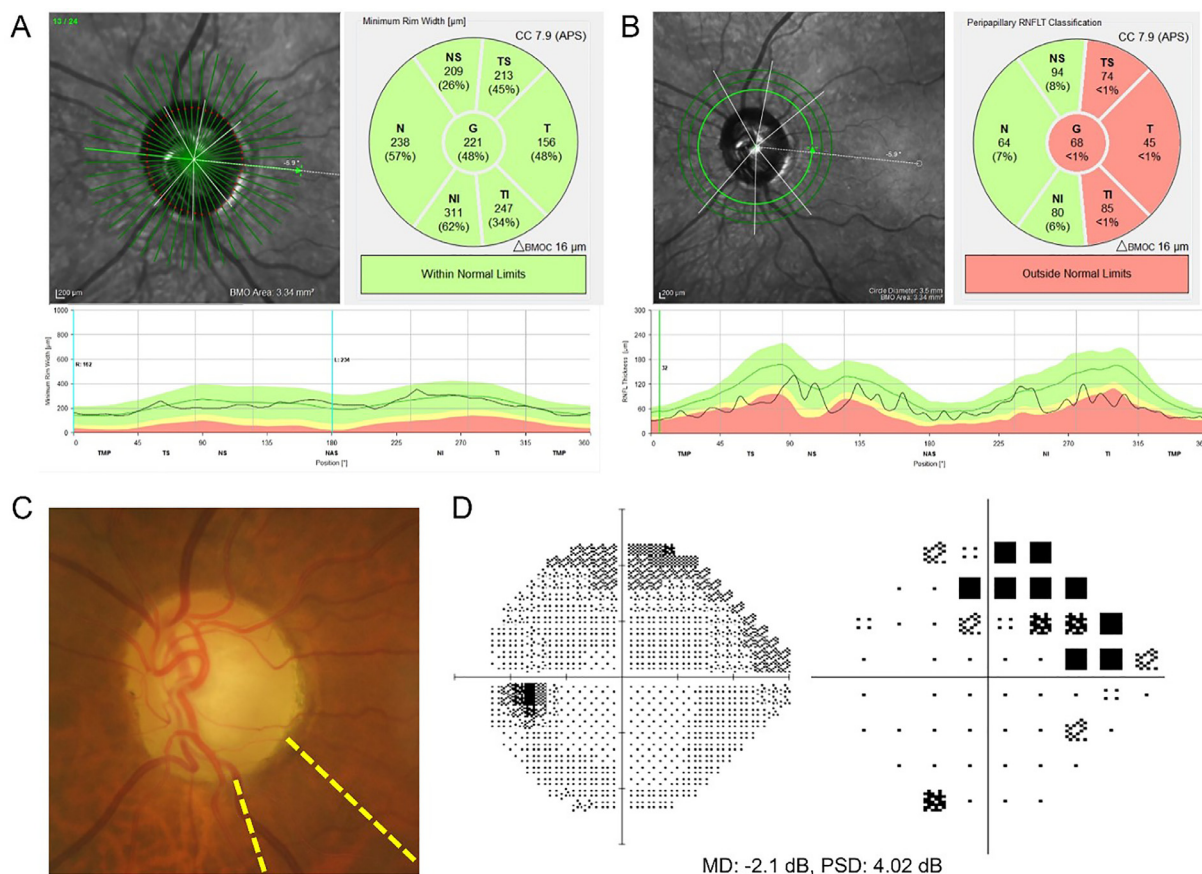


FIGURE 2. Case example of false negative for Bruch membrane opening minimum rim width. Optical coherence tomography scans from a 77-year-old male glaucoma patient of African descent with a Bruch membrane opening area of 3.34 mm². (A) Global and sectoral Bruch membrane opening minimum rim width were within normal limits. (B) Global and sectoral (temporal, superotemporal, and inferotemporal) retinal nerve fiber layer thicknesses were below the 1st percentile of the normal reference values. (C) Retinal nerve fiber layer defect was found in the inferotemporal sector (yellow dashed lines). (D) Visual field result showing corresponding partial arcuate defect. MD = mean deviation, PSD = pattern standard deviation.

In ED participants, unlike the results for AD participants, integrating the combination of RNFL or BMO-MRW measurements as the diagnostic criterion increased the sensitivity of RNFLT (all: 92.5% to 95.3%, $P = .025$; early: 87.3% to 92.5%, $P = .001$) and BMO-MRW (all: 88.5% to 95.3%, $P = .008$; early: 82.1% to 92.5%, $P = .001$) assessment without compromising their specificities for glaucoma (RNFLT: $P = .360$, and BMO-MRW: $P = .508$) or early glaucoma (RNFLT: $P = .461$, and BMO-MRW: $P = .579$).

DISCUSSION

In this study, we compared the sensitivities and specificities of RNFLT and BMO-MRW measurements of 6 different classification criteria to interpret the diagnostic classifications of OCT used by clinicians for the detection of

glaucoma in individuals of African and European descents. Overall, sensitivities and specificities of all diagnostic criteria were significantly lower in AD than in ED participants, whereas choice of the criteria to apply depends on the relative importance of sensitivity and specificity in specific patient populations and settings. Specifically, the interpretation of FN results in BMO-MRW printout should be done with caution in AD participants as up to one-fourth of the patients might have normal OCT. ST and/or IT RNFLT below the 5th percentile has the best performance for detection of glaucoma among RNFLT classifications with higher sensitivity but lower specificity than that of ST and/or IT BMO-MRW below the 5th percentile in AD and ED participants. The results of the present study suggest that improvements in the reference database, perhaps by implementing race-specific databases,^{34,38} are needed to assess whether OCT measurements are outside normal limits in AD patients, especially when using optic nerve head rim parameters.

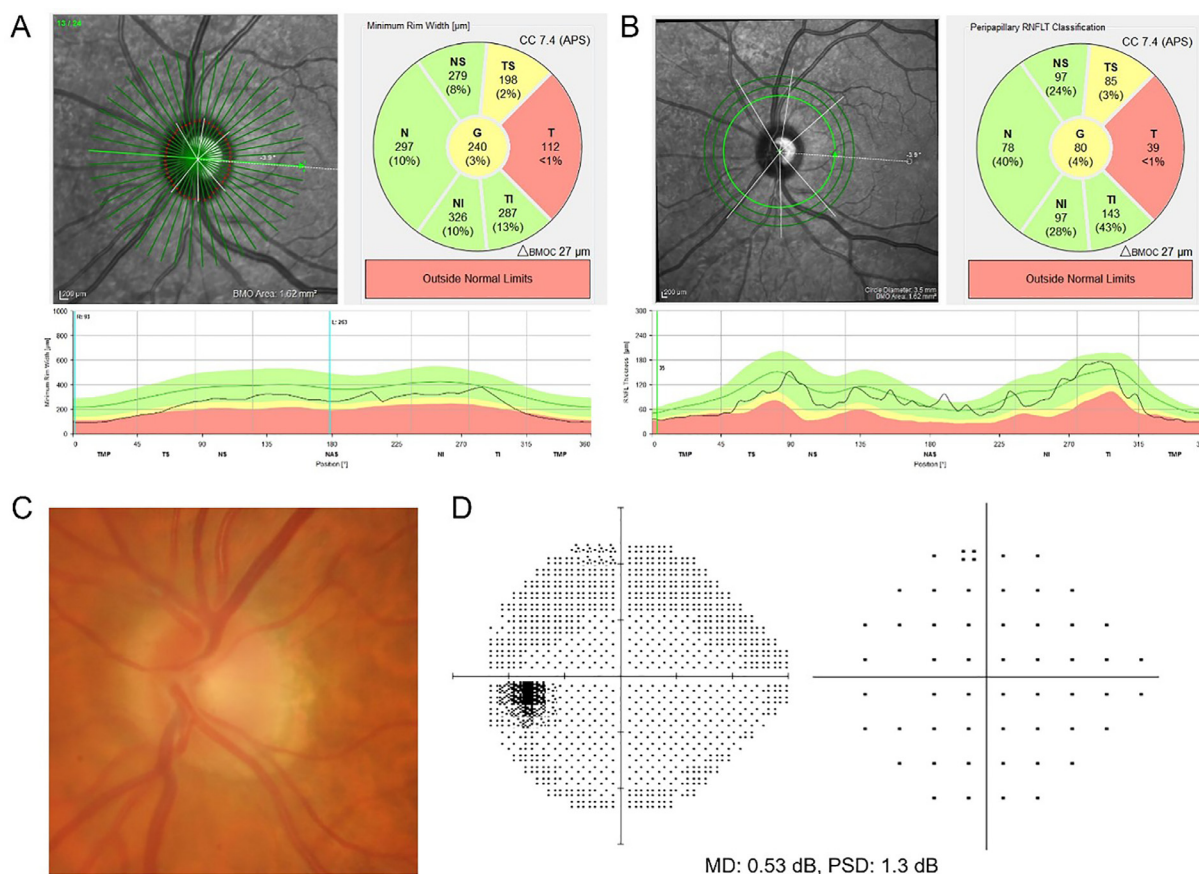


FIGURE 3. Case example of healthy participants below the 5th percentile for both Bruch membrane opening minimum rim width and retinal nerve fiber layer thickness. OCT scans from a 51-year-old female healthy participant of African descent with a Bruch membrane opening area of 1.62 mm². (A) Global Bruch membrane opening minimum rim width was below the 5th percentile of the normal reference values. (B) Global retinal nerve fiber layer thickness was below the 5th percentile of the normal reference values. (C) Disc photo showing normal optic disc appearance. (D) Visual field result showing no abnormality. MD = mean deviation, PSD = pattern standard deviation.

Although the detection of neuroretinal rim narrowing and RNFL thinning often relies on the diagnostic classification reports of OCT in clinical practice, our understanding of the RNFLT and BMO-MRW diagnostic performance is heavily dependent on the AUROC curve analysis related to the global or sectoral RNFLT and BMO-MRW assessments.^{25–27,33} Many studies have found no differences in the diagnostic performance between global RNFLT and BMO-MRW. For instance, one study compared the differences in AUROC curve analysis between RNFLT and BMO-MRW in preperimetric and perimetric glaucoma patients and showed a similar AUC between global RNFLT and BMO-MRW for detecting glaucoma in these patients.²² Another study evaluated the diagnostic performance of the global measurements in myopic glaucoma patients and healthy controls and concluded that global RNFLT and BMO-MRW are comparable for the detection of glaucoma in myopic eyes.

In our study, we evaluated the RNFLT and BMO-MRW diagnostic classification reports and observed higher sensi-

tivities of the global RNFLT than the global BMO-MRW, whereas specificities of BMO-MRW tended to be higher than for RNFLT for all participants. Zheng and colleagues analyzed the diagnostic classification reports of RNFLT and BMO-MRW and found that global RNFLT had higher sensitivity (88.8%) and similar specificity (98.7%) compared with global BMO-MRW (sensitivity: 78.7%, specificity: 99.3%),²⁸ which partially agrees with our findings, possibly because of the differences in measurements across ethnicities (Chinese vs AD and ED).

The disagreement between the results using AUROC analysis reported by previous studies and the diagnostic classification report is that the sensitivities were compared at a lower specificity (90%-95%) using the AUROC analysis, whereas the sensitivities of global RNFLT and BMO-MRW below the 1st and the 5th percentile derived from the diagnostic classification report were compared at higher specificities (95.7%-100.0%). The specificities were higher in the latter method because RNFLT and BMO-MRW abnormalities are defined though

TABLE 4. Sensitivities and Specificities of RNFLT and BMO-MRW for Detection of Early Glaucoma in AD and ED Participants

	Race	RNFLT	BMO-MRW	Combined RNFLT or BMO-MRW	Difference (RNFLT-BMO MRW)	P Value*
1. Global measurement below the 5th percentile						
Sensitivity	AD	61.4 (48.6, 75.0)	44.3 (30.5, 58.8)	70.0 (56.3, 82.8)	17.1 (2.2, 33.9)	.034
	ED	64.2 (54.9, 72.0)	56.7 (47.5, 65.4)	73.1 (63.9, 80.4)	7.5 (−0.8, 16.9)	.106
Specificity	AD	94.7 (87.2, 100.0)	97.4 (90.2, 100.0)	94.7 (87.2, 100.0)	−2.6 (−11.1, 0.0)	.326
	ED	96.4 (90.4, 100.0)	100.0 (100.0, 100.0)	96.4 (90.4, 100.0)	−3.6 (−9.6, 0.0)	.155
2. Global measurement below the 1st percentile						
Sensitivity	AD	45.7 (32.9, 60.3)	17.1 (8.2, 30.5)	48.6 (36.2, 63.1)	28.6 (13.6, 44.2)	<.001
	ED	44.8 (34.9, 53.5)	32.8 (24.0, 41.2)	50.7 (41.3, 59.6)	11.9 (3.6, 20.3)	.005
Specificity	AD	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—
	ED	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (0.0, 0.0)	—
3. One or more sectoral measures below the 5th percentile						
Sensitivity	AD	78.6 (67.9, 88.1)	65.7 (52.9, 77.8)	87.1 (79.1, 94.3)	12.9 (−2.3, 26.0)	.084
	ED	89.6 (83.7, 94.2)	88.1 (81.9, 93.1)	96.3 (92.5, 99.2)	1.5 (−6.1, 8.2)	.687
Specificity	AD	65.8 (45.9, 85.0)	86.8 (71.8, 100.0)	63.2 (42.9, 80.6)	−21.1 (−40.0, −3.2)	.029
	ED	85.7 (73.2, 94.1)	94.6 (87.7, 100.0)	82.1 (69.3, 92.4)	−8.9 (−22.0, 0.0)	.097
4. One or more sectoral measures below the 1st percentile						
Sensitivity	AD	65.7 (53.6, 77.3)	44.3 (29.6, 58.5)	74.3 (61.9, 85.5)	21.4 (6.9, 38.3)	.008
	ED	73.9 (66.5, 80.5)	70.1 (61.5, 79.1)	85.8 (78.9, 91.7)	3.7 (−6.1, 12.3)	.425
Specificity	AD	84.2 (66.7, 96.9)	97.4 (90.2, 100.0)	84.2 (66.7, 96.9)	−13.2 (−28.2, −2.7)	.038
	ED	98.2 (92.4, 100.0)	100.0 (100.0, 100.0)	98.2 (92.4, 100.0)	−1.8 (−7.6, 0.0)	.350
5. Superotemporal and/or inferotemporal measurement below the 5th percentile						
Sensitivity	AD	71.4 (59.1, 82.8)	57.1 (43.1, 68.9)	78.6 (67.2, 88.7)	14.3 (0.0, 29.5)	.068
	ED	87.3 (81.5, 92.9)	82.1 (74.6, 88.8)	92.5 (87.0, 96.6)	5.2 (−2.0, 12.7)	.163
Specificity	AD	73.7 (53.4, 91.4)	89.5 (74.4, 100.0)	71.1 (50.0, 90.0)	−15.8 (−38.0, 0.0)	.094
	ED	96.4 (90.0, 100.0)	98.2 (93.2, 100.0)	94.6 (87.6, 100.0)	−1.8 (−8.5, 4.2)	.574
6. Superotemporal and/or inferotemporal measurement below the 1st percentile						
Sensitivity	AD	64.3 (52.7, 76.9)	31.4 (19.1, 45.0)	67.1 (55.2, 79.4)	32.9 (19.7, 48.0)	<.001
	ED	71.6 (64.9, 79.0)	57.5 (48.4, 66.9)	79.9 (72.5, 87.0)	14.2 (4.9, 22.7)	.002
Specificity	AD	92.1 (76.0, 100.0)	100.0 (100.0, 100.0)	92.1 (76.0, 100.0)	−7.9 (−24.0, 0.0)	.173
	ED	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	100.0 (100.0, 100.0)	0.0 (−0.0, 0.0)	—

AD = African descent, BMO-MRW = Bruch membrane opening minimum rim width, ED = European descent, RNFLT = retinal nerve fiber layer thickness. Estimates of sensitivity and specificity (%) of RNFLT and BMO-MRW (95% CI) for early glaucoma subjects vs healthy controls for each diagnostic criteria including African descent (AD) and European descent (ED) participants.

*P value for difference between RNFLT and BMO-MRW sensitivity and specificity. Statistical significance: $P < .05$.

the use of the 1st and 5th percentile of the reference data.

Despite studies suggesting differences in the optic nerve parameters based on ethnicity, it is not standard practice to account for race when determining normal ranges of OCT measurements: however, the Spectralis has a Japanese-specific database, with a Hispanic/Latino and African descent reference database under development. Spectralis does adjust for optic disc size, which has explained at least in part the higher risk of developing glaucoma in ocular hypertensive AD individuals in the Ocular Hypertension Treat-

ment Study.²² However, adjustment for optic disc size did not adequately account for differences between the diagnostic accuracy of RNFLT and BMO-MRW in our AD and ED participants.

In the present study, ST and/or IT RNFLT below the 5th percentile maintained the best performance for detection of glaucoma and early glaucoma in AD and ED participants with lower sensitivities and specificities of RNFLT and BMO-MRW in AD participants compared with ED participants. These findings may be related to the racial

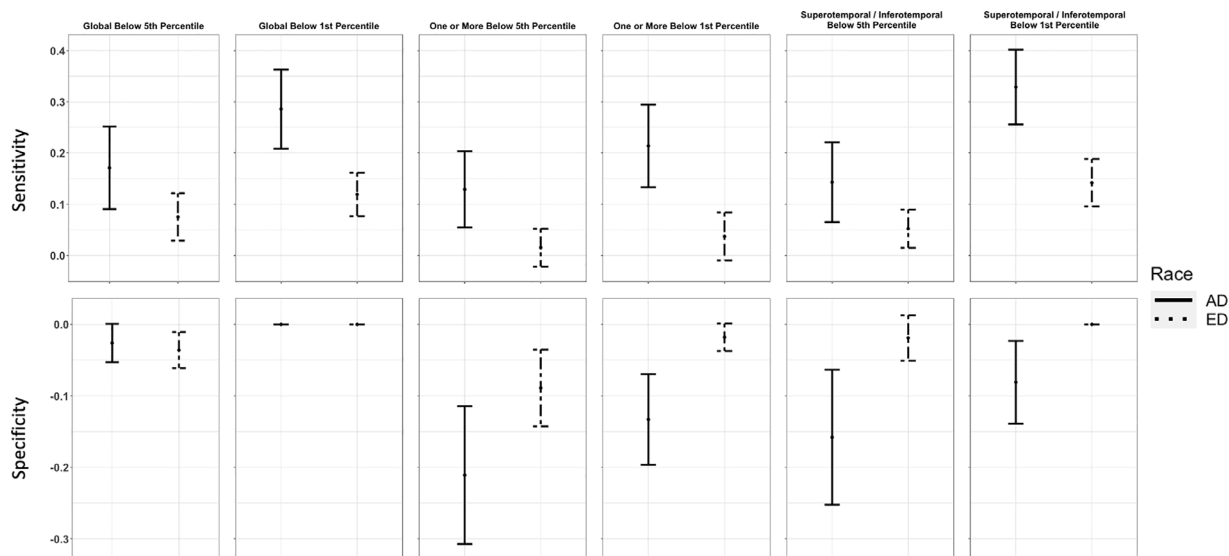


FIGURE 4. Sensitivities and specificities of RNFLT and BMO-MRW for detection of early glaucoma in AD and ED participants. Error bars represent the 95% CI. AD = African descent, BMO-MRW = Bruch membrane opening minimum rim width, ED = European descent, RNFLT = retinal nerve fiber layer thickness.

differences that exist in the ONH, RNFL, and BMO-MRW measurements.^{39–44} In our study, we relied on the newer age and optic disc size adjusted reference database for GMPE, which includes 368 normal participants with various racial and ethnic backgrounds, including 246 Caucasian participants, 47 Hispanics, 47 Black or African Americans, and 25 Asians.

Although the newer reference database used in this study reflects the US population better than the prior reference database, which included mainly Caucasians with very few participants of African and Asian descents, it may not be diverse enough to adequately represent the normal range of population variations in the RNFL and BMO-MRW measurements. Given these racial differences, it is critical to examine the accuracy of the current GMPE reference database for classifying populations with diverse ethnic backgrounds.

A recent study evaluated the differences in RNFLT between AD and the reference database of the Cirrus HD-OCT and suggested that the existing reference RNFLT database in the Cirrus HD-OCT is likely to overdiagnose ONH abnormality in the temporal quadrant in AD patients.²⁹ Specifically, the thinner temporal quadrant in AD produced more abnormal results (8.2%) than expected (5.0%) by the Cirrus HD-OCT.³⁴ Multiple studies also evaluated the reference database for Asian populations, noting some variability between Asians (ie, thicker RNFL in the superior, inferior, and temporal quadrants) and the reference database of various OCTs.^{38,45–50} For instance, one study found that when the Cirrus HD-OCT outside normal limits determinations were adjusted based on healthy Asian participants, 4% of the normal Cirrus RNFL measurements were recoded as abnormal.⁴⁹ Therefore, it is important to consider establishing a race-specific reference database to

improve the sensitivity and specificity of RNFLT and BMO-MRW in certain populations.

In our study, BMO-MRW had a higher FN rate than RNFL, which was worse in AD (up to 28%) compared with ED. This finding may be related to the greater optic disc area, larger cup-to-disc ratio, and bigger BMO area in AD. Our results agree with previous reports of FN rates in early glaucoma patients, where they found a higher number of FNs for BMO-MRW (20/32) compared with RNFL (12/32).⁵¹ Another study found that large optic disc and myopia might show abnormal RNFL classifications but normal classifications based on BMO-MRW. However, the normal VF in most of their cases (86.1%) led to the interpretation that it is likely an FP outside normal limits classification of RNFL than FN color code classification of BMO-MRW.

It is also worth mentioning that it could be cases of preperimetric glaucoma.³² FN rate of BMO-MRW in perimetric glaucoma is high and would likely be higher in preperimetric glaucoma patients of AD. Although race-specific reference databases are under development, it remains to be determined whether sensitivity and specificity will improve with their use, as there is considerable variability within racial and ethnic groups. Alternatively, as there are optic disc characteristics such as disc size, spherical equivalent, axial length, or the relationship of the anterior scleral canal opening to BMO,⁵² blood vessel locations,⁵¹ and other systemic conditions³⁸ that are associated with RNFLT, it may be more effective to adjust for these factors and increase their ranges (eg, range of optic disc sizes) than race and ethnicity to improve the accuracy of detecting eyes with glaucoma across diverse populations.

We investigated 6 diagnostic criteria of RNFLT and BMO-MRW for the diagnostic evaluation of glaucoma and early glaucoma and observed a higher sensitivity of RNFLT than BMO-MRW in global and sectoral measurements. Although choice of the criteria to apply depends on the relative importance of sensitivity and specificity in specific patient populations and settings, among the 6 diagnostic criteria, ST and/or IT RNFLT below the 5th percentile had the best performance for detection of glaucoma among RNFLT classifications with a significantly higher sensitivity but lower specificity than that of ST and/or IT BMO-MRW below the 5th percentile in all participants. It is expected for the ST and/or IT measurement below the 5th percentile to show the best performance for the detection of glaucoma because these sectors are the most common locations for glaucoma to develop and progress.^{53–56}

The better sensitivity of RNFLT compared with BMO-MRW, which may be more important than specificity in high-risk patients like African Americans, suggests that RNFLT is a better marker than BMO-MRW in detecting glaucoma and early glaucoma in these patients. A systemic review showed that inferior RNFLT was the best-performing parameter with pooled sensitivity and specificity of 80% and 94%, respectively.⁵⁷ The wide variations in the diagnostic performance of OCT RNFLT and BMO-MRW parameters may be due to variation in the definition of glaucoma used in the analysis; severity of glaucoma in the study population; and by different characteristics, such as age, gender, race, and refractive error, of the populations enrolled in each study.

Our study has several limitations. First, our sample size is relatively small and may limit our ability to detect significant differences, particularly in the AD participants and healthy controls. Second, the high sensitivities observed in the current study are possibly related to the definition of glaucoma based on the severity of the VF damage since

glaucomatous eyes with VF damage are likely to have RNFLT loss. Third, we did not include eyes with AL > 26.0 mm given that the Spectralis OCT does not include reference database from highly myopic eyes, so that our results are not generalizable to myopic individuals.⁵⁸ Fourth, ROC curves can provide additive comprehensive information when compared to sensitivities and specificities. Analysis of ROC curves will be the aim of future studies while controlling for confounders. Fifth, although we excluded patients with diabetic retinopathy, participants with controlled diabetes were included in the analysis, and there was a higher prevalence of diabetes in both the healthy and glaucoma AD population compared to the ED population. This may lead to some differences between the groups because of the reported association between diabetes and RNFLT loss.⁵⁹ Last, our data mainly came from African Americans, which may not be reflective of all populations of AD.

In conclusion, RNFLT reference database criteria for detecting glaucoma or early glaucoma had higher sensitivity, but lower specificity, than BMO-MRW classifications in both AD and ED individuals. RNFLT and BMO-MRW measurements had consistently lower diagnostic performance in AD individuals than in ED individuals. Moreover, BMO-MRW criteria might fail to detect as many as one-third of eyes with glaucoma, particularly in AD individuals. It is notable that the BMO-MRW printout can be normal in one-fourth of these glaucoma AD patients. Therefore, clinicians should be careful when using OCT with the current Spectralis reference database because RNFLT, and especially BMO-MRW, criteria are not adequate for diagnosing glaucoma in AD individuals, which may lead to misidentification of some glaucomatous eyes and failure to detect early glaucomatous damage in such individuals. A large ethnicity-specific reference database is needed to assess OCT color probability codes in AD patients.

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